AMAL JYOTHI COLLEGE OF ENGINEERING
KANJIRAPPALLY

B. TECH - COMPUTER SCIENCE AND ENGINEERING

BTCS2023-27-S6: 23CST362-PROGRAMMING IN PYTHON-
Project (Batch – I)

FRAUD TRANSACTION DETECTION SYSTEM

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1. Abstract

The rapid growth of digital payment systems and online banking has increased the risk of financial fraud such as unauthorized transactions and payment scams. Detecting fraudulent activities manually is difficult due to the large volume of transactions processed daily. This project presents a Python-based Fraud Transaction Detection System that automatically identifies suspicious transactions using machine learning techniques. The system uses the Logistic Regression algorithm from the Scikit-learn library to analyze transaction features such as transaction time and transaction amount. A graphical user interface developed using Tkinter allows users to manually check transactions and simulate live transaction monitoring. Data visualization using Matplotlib helps represent fraud statistics clearly. The system demonstrates how Python libraries can be used to build a simple fraud detection solution for financial security applications and digital transaction monitoring.

2. Problem Description

In modern financial systems, millions of digital transactions occur every day through credit cards, online banking platforms, and digital payment applications. Although these technologies make financial transactions faster and more convenient, they also create opportunities for fraudulent activities such as unauthorized payments, identity theft, and transaction manipulation.

Financial institutions must identify fraudulent transactions quickly to prevent financial losses and protect customers. Manual monitoring of transactions is not efficient because of the enormous volume of transactions processed every day.

Automated fraud detection systems using machine learning techniques can analyze transaction patterns and identify unusual behavior that may indicate fraudulent activity. These systems learn from historical transaction data and detect anomalies in new transactions.

The objective of this project is to design and implement a Python-based Fraud Transaction Detection System that can analyze transaction data and classify transactions as legitimate or fraudulent. The system also provides an interactive

graphical interface and live monitoring functionality to simulate real-world fraud detection systems used in financial institutions.

3. Libraries and Tools Used

This project uses several Python libraries to implement the fraud detection system.

Pandas

Pandas is a data analysis library used for loading, processing, and managing structured data. In this project, Pandas is used to read the credit card transaction dataset and prepare it for machine learning analysis.

Scikit-learn

Scikit-learn is a machine learning library that provides tools for building predictive models. In this project, the Logistic Regression algorithm from Scikit-learn is used to train a fraud detection model.

Tkinter

Tkinter is Python's built-in library for creating graphical user interfaces. It is used to design the application's user interface, which allows users to interact with the fraud detection system.

Matplotlib

Matplotlib is a data visualization library used to create graphs and charts. In this project, it is used to generate a pie chart that visualizes the proportion of fraudulent and legitimate transactions.

Python

Python is the programming language used to implement the entire project due to its simplicity, flexibility, and strong support for machine learning and data analysis libraries.

4. System Design / Workflow

The Fraud Transaction Detection System follows a simple workflow consisting of multiple stages.

Step 1: Dataset Loading

The system loads the credit card transaction dataset using the Pandas library. The dataset contains information about transaction attributes and a label indicating whether a transaction is fraudulent.

Step 2: Data Preprocessing

The dataset is divided into input features and the target variable. Input features include transaction characteristics such as time and amount, while the target variable indicates whether the transaction is fraudulent.

Step 3: Model Training

A machine learning model using Logistic Regression is trained using the dataset. The dataset is divided into training and testing sets to build an effective classification model.

Step 4: Transaction Input

Users can enter transaction details such as transaction time and transaction amount using the graphical interface.

Step 5: Fraud Prediction

The trained machine learning model analyzes the input transaction and calculates the probability that the transaction is fraudulent.

Step 6: Result Display

The system displays the fraud risk score and categorizes the transaction as legitimate, suspicious, or fraudulent.

Step 7: Dashboard Monitoring

The application maintains a dashboard that displays important statistics including the total number of transactions analyzed, number of frauds detected, and overall fraud rate.

Step 8: Live Monitoring

The system can simulate real-time transaction monitoring by automatically analyzing randomly generated transactions at regular intervals.

5. Implementation Details

The Fraud Transaction Detection System is implemented using Python and several external libraries.

Dataset Handling

The dataset is loaded using the Pandas library and stored in a DataFrame structure. The dataset includes transaction attributes and a class label representing fraud status.

Machine Learning Model

A Logistic Regression model is used to detect fraudulent transactions. The dataset is split into training and testing sets using the train_test_split function from Scikit-learn.

Fraud Risk Calculation

The trained model predicts the probability that a given transaction is fraudulent. This probability is converted into a fraud risk score.

Transaction Classification

Based on the calculated risk score, transactions are classified as:

- Legitimate Transaction
- Suspicious Transaction
- Fraudulent Transaction

Graphical User Interface

The graphical interface is built using Tkinter. The interface includes:

- Dataset loading button
- Model training button
- Transaction input fields
- Fraud detection result display
- Transaction monitoring table
- Fraud statistics dashboard

Visualization

Matplotlib is used to generate a pie chart showing the distribution of legitimate and fraudulent transactions in the dataset.

Live Monitoring

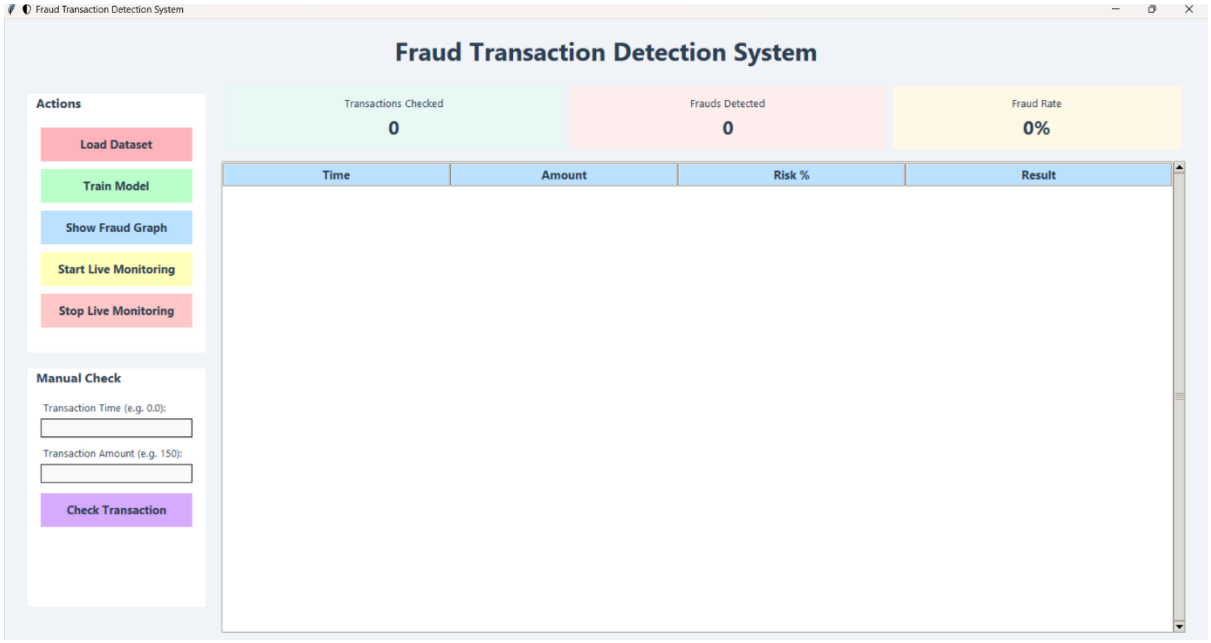
The system includes a live monitoring feature that automatically processes random transactions and displays the results in the transaction table.

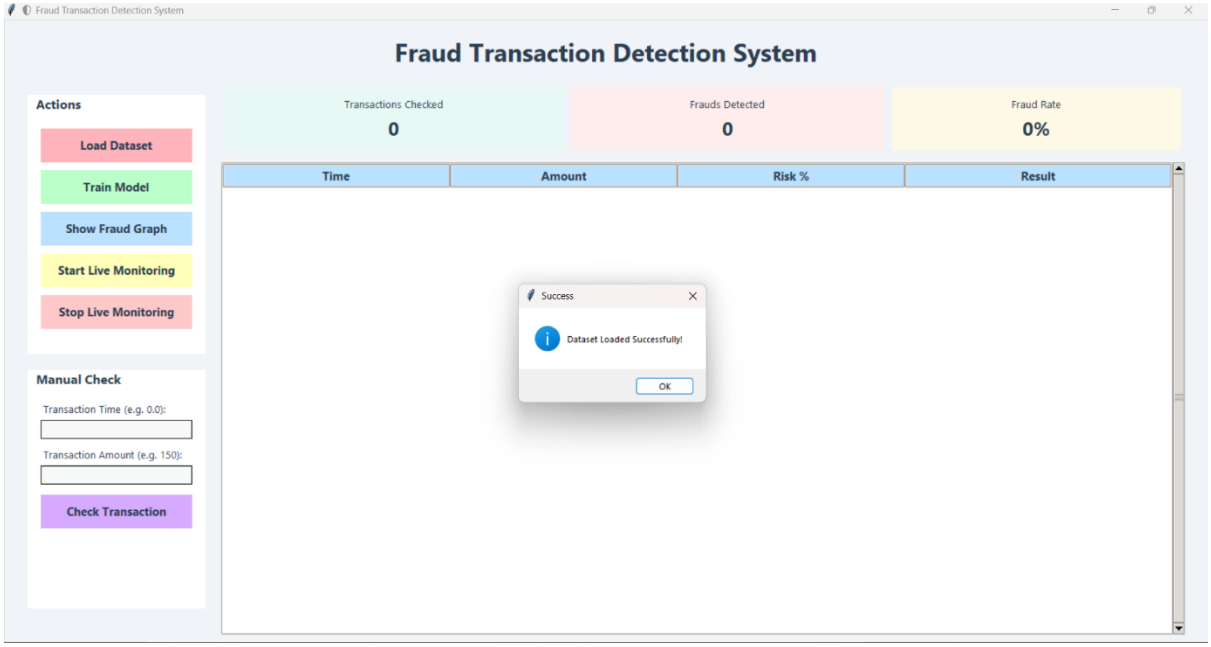
6. Results and Output

The system successfully analyses transaction data and identifies potentially fraudulent activities.

Application Interface

The graphical interface provides buttons for loading datasets, training the model, and analyzing transactions.


Fraud Transaction Detection System
Fraud Transaction Detection System
Actions
Load Dataset
Train Model
Show Fraud Graph
Start Live Monitoring
Stop Live Monitoring
Manual Check
Transaction Time (e.g. 0.0):
Transaction Amount (e.g. 150):
Check Transaction
Transactions Checked
0
Frauds Detected
0
Fraud Rate
0%
Time
Amount
Risk %
Result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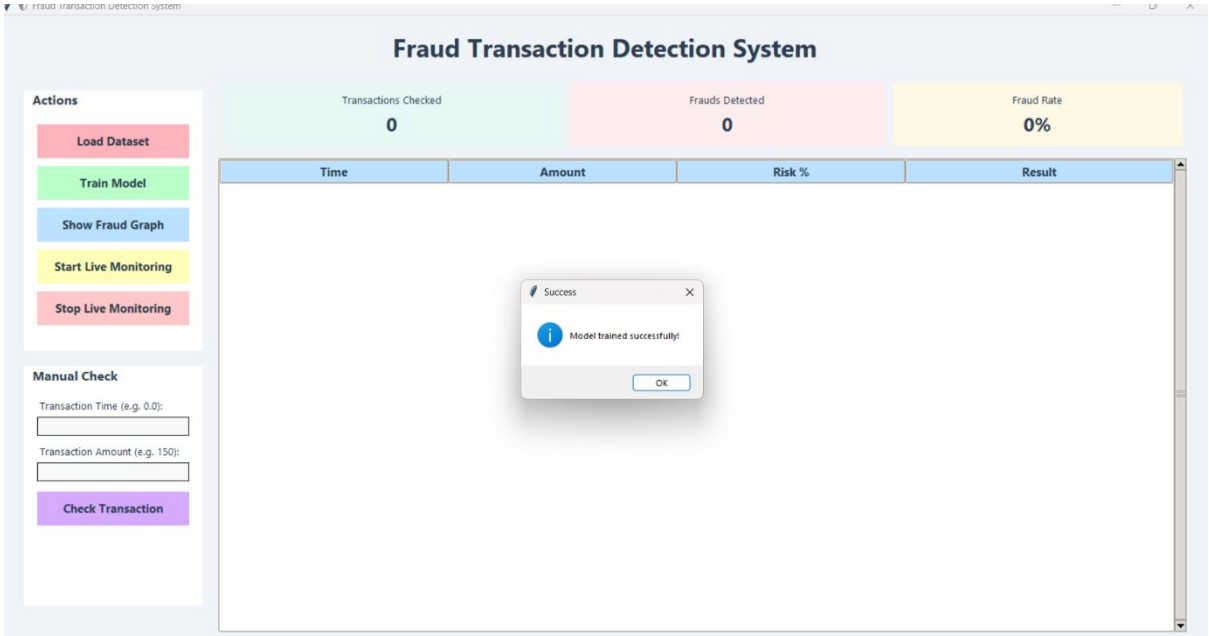
Check Transaction
Transactions Checked
0
Frauds Detected
0
Fraud Rate
0%
Time
Amount
Risk %
Result
Success
Dataset Loaded Successfully!
OK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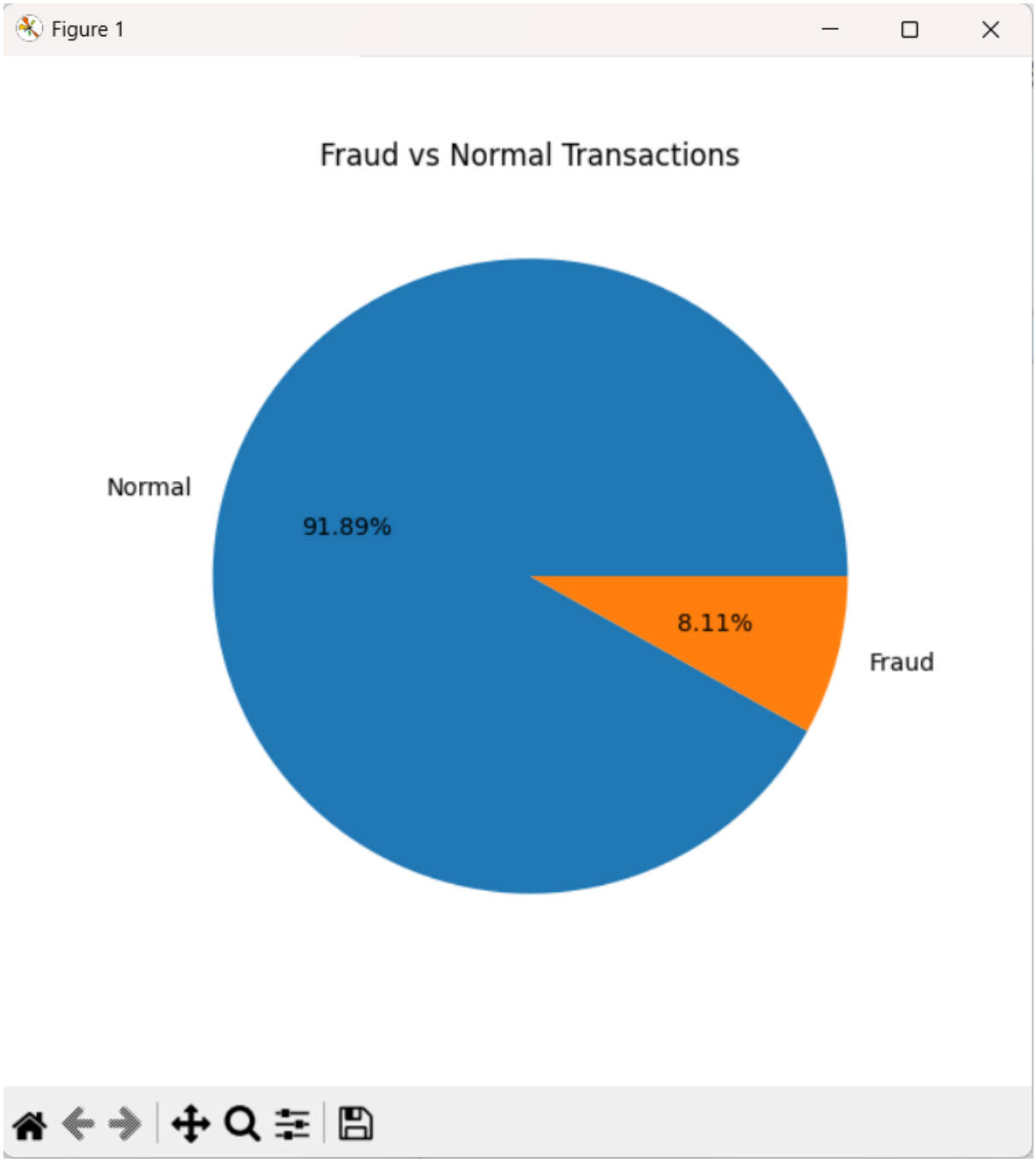
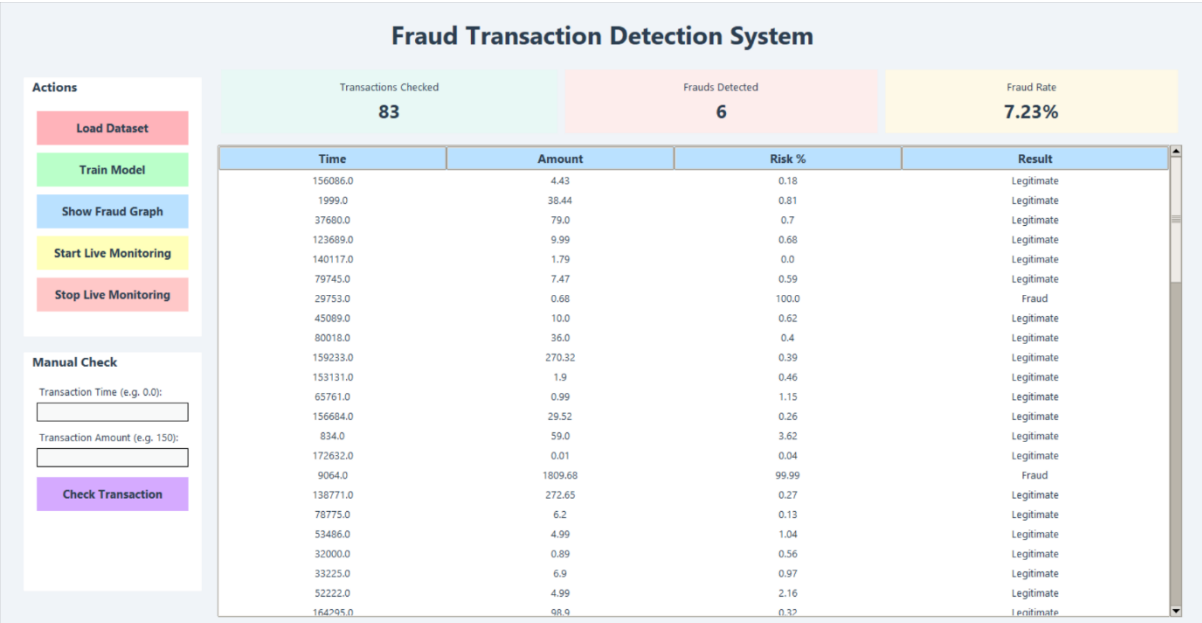
Check Transaction
Transactions Checked
0
Frauds Detected
0
Fraud Rate
0%
Time
Amount
Risk %
Result
Success
Model trained successfully!
OK



Live Monitoring

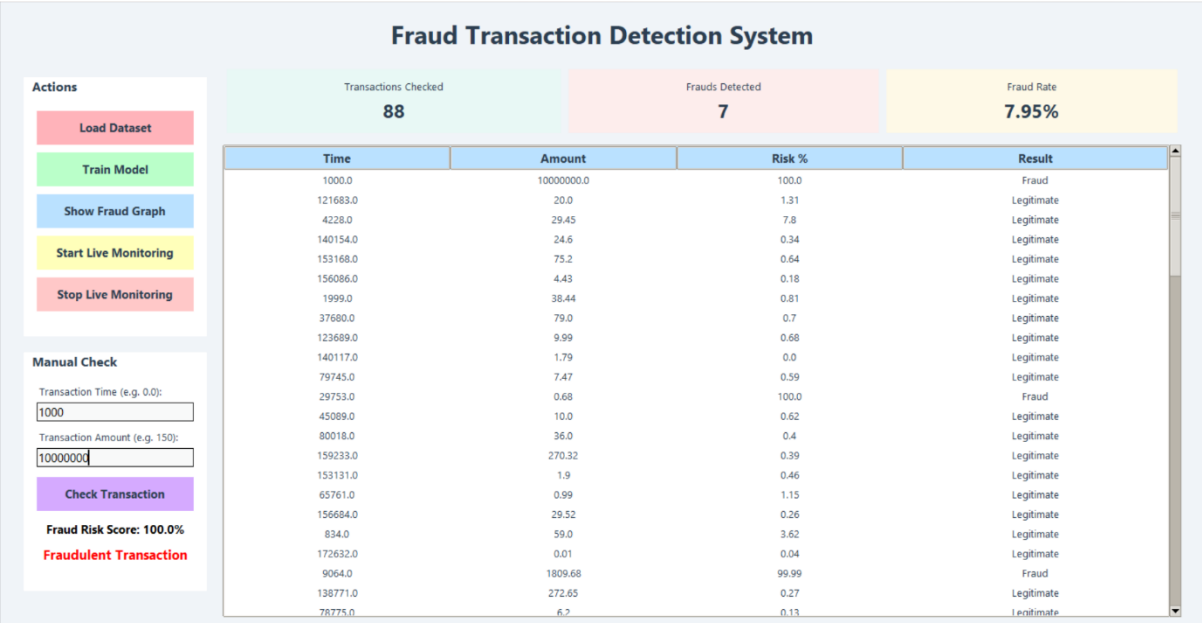
The live monitoring feature simulates continuous transaction analysis and updates the dashboard automatically.



Time	Amount	Risk %	Result
156086.0	4.43	0.18	Legitimate
1999.0	38.44	0.81	Legitimate
37680.0	79.0	0.7	Legitimate
123689.0	9.99	0.68	Legitimate
140117.0	1.79	0.0	Legitimate
79745.0	7.47	0.59	Legitimate
29753.0	0.68	100.0	Fraud
45089.0	10.0	0.62	Legitimate
80018.0	36.0	0.4	Legitimate
159233.0	270.32	0.39	Legitimate
153131.0	1.9	0.46	Legitimate
65761.0	0.99	1.15	Legitimate
156684.0	29.52	0.26	Legitimate
834.0	59.0	3.62	Legitimate
172632.0	0.01	0.04	Legitimate
9064.0	1809.68	99.99	Fraud
138771.0	272.65	0.27	Legitimate
78775.0	6.2	0.13	Legitimate
53486.0	4.99	1.04	Legitimate
32000.0	0.89	0.56	Legitimate
33225.0	6.9	0.97	Legitimate
52222.0	4.99	2.16	Legitimate
164295.0	98.9	0.32	Legitimate

Fraud Detection Result

Users can manually enter transaction values and obtain the fraud risk score and classification result.



Time	Amount	Risk %	Result
1000.0	10000000.0	100.0	Fraud
121683.0	20.0	1.31	Legitimate
4228.0	29.45	7.8	Legitimate
140154.0	24.6	0.34	Legitimate
153168.0	75.2	0.64	Legitimate
156086.0	4.43	0.18	Legitimate
1999.0	38.44	0.81	Legitimate
37680.0	79.0	0.7	Legitimate
123689.0	9.99	0.68	Legitimate
140117.0	1.79	0.0	Legitimate
79745.0	7.47	0.59	Legitimate
29753.0	0.68	100.0	Fraud
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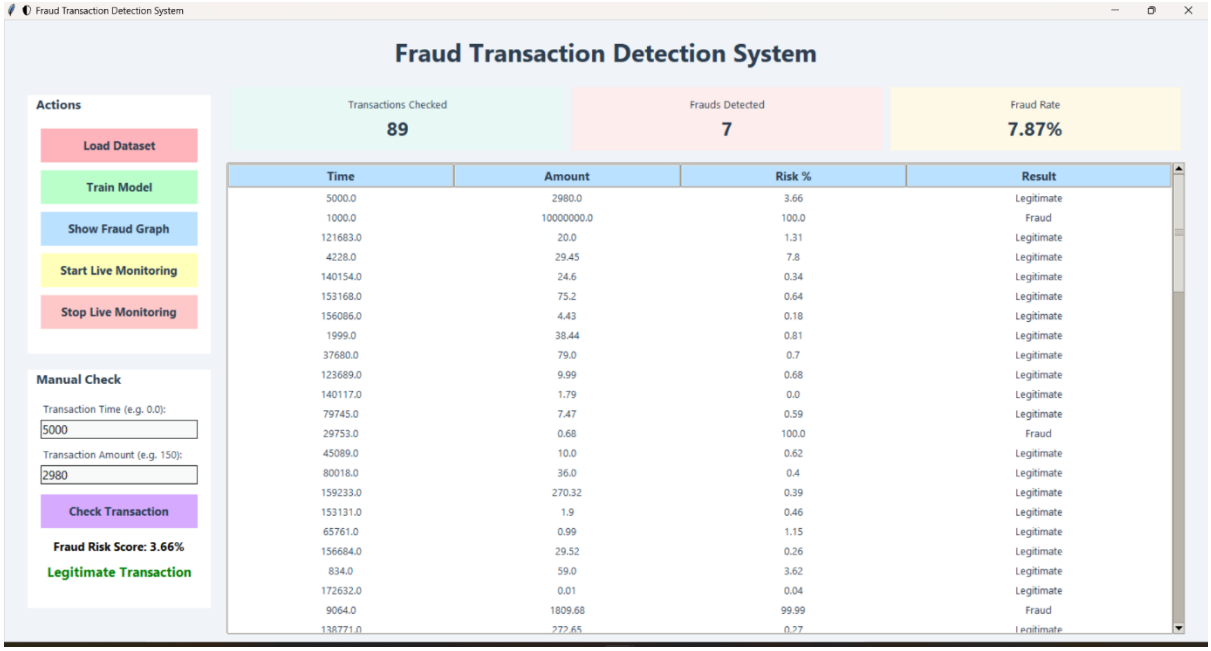

The results demonstrate that the system can effectively detect suspicious transactions using machine learning techniques.

7. Conclusion

The Fraud Transaction Detection System demonstrates how Python and machine learning techniques can be applied to detect suspicious financial transactions. The system integrates data analysis, machine learning, and graphical user interface design into a single application.

The developed system can classify transactions as legitimate, suspicious, or fraudulent based on calculated risk scores. The live monitoring feature simulates real-time fraud detection similar to systems used in modern banking environments.

This project highlights the importance of automated fraud detection systems in improving financial security and preventing financial losses.

8. Scope for Mini Project Extension

The current system can be further enhanced in several ways to improve its functionality and performance.

Possible improvements include:

- Integration with real-time banking transaction APIs
- Implementation of advanced machine learning models such as Random Forest or Neural Networks
- Development of a web-based fraud detection platform
- Integration with cloud-based monitoring systems
- Addition of advanced anomaly detection algorithms

These improvements would allow the system to be used in real-world financial applications.

9. References

Scikit-learn Documentation – https://scikit-learn.org

Pandas Documentation – https://pandas.pydata.org

Matplotlib Documentation – https://matplotlib.org

Credit Card Fraud Detection Dataset – Kaggle